DGMS 2019 • SECTION J: ECONOMIC & PETROLEUM GEOLOGY

Question 78: Indian Hydrocarbon & Fossil Fuel Deposits Matching

Official Examination Guide • Geographic Distribution & Stratigraphy of Indian Energy Resources

OFFICIAL EXAMINATION QUESTION

Match List-I (Hydrocarbon deposit) with List-II (Location):

LIST-I (DEPOSIT)	LIST-II (LOCATION)
A. Lignite	**1.** Kalol, Gujarat
B. Oil	**2.** Neyveli, Tamil Nadu
C. Bituminous coal	**3.** Jaisalmer, Rajasthan
D. Natural gas	**4.** Kamte (Kamptee), Maharashtra

(a) A-2, B-1, C-4, D-3 CORRECT ANSWER

(b) A-1, B-2, C-3, D-4

(c) A-2, B-4, C-3, D-1

(d) A-3, B-1, C-4, D-2

1. Executive Summary & Verification

India's energy resource distribution spans distinctive sedimentary basins across various geological eras—from Lower Gondwana Permian coal measures to Cenozoic rift-margin petroleum basins and Tertiary coastal lignites:

- **A → 2: Lignite in Neyveli, Tamil Nadu:** Neyveli hosts India's premier Tertiary lignite deposit within the Miocene Cuddalore Sandstone Formation

in the Cauvery Basin / Tamil Nadu coastal plain. Mined by NLC India Ltd., it supplies fuel to giant pit-head thermal stations.

- **B → 1: Oil in Kalol, Gujarat:** Kalol is a prolific onshore oilfield situated in the Ahmedabad-Mehsana block of the Cambay Basin, discovered by the Oil and Natural Gas Corporation (ONGC) in June 1961. The Middle Eocene Kalol Formation contains multiple deltaic sandstone reservoir pays.
- **C → 4: Bituminous Coal in Kamptee (Kamte), Maharashtra:** The Kamptee Coalfield in Nagpur district belongs to the Wardha-Kanhan-Godavari valley Lower Gondwana basin. The Permian Barakar and Motur/ Kamthi formations host substantial reserves of non-coking bituminous coal, extracted by Western Coalfields Limited (WCL).
- **D → 3: Natural Gas in Jaisalmer, Rajasthan:** The Jaisalmer Basin on the northwestern Indian shelf is a premier gas-producing basin. Discovered by OIL and ONGC, major gas fields include Manhera Tibba, Tanot, Dandewala, and Ghotaru, producing commercial gas from Cretaceous (Pariwar) and Tertiary (Sanu/Bandah) sandstone/limestone reservoirs.

Combining these unambiguous pairs yields the unique matching sequence **A-2, B-1, C-4, D-3**, which corresponds precisely to Option **(a)**.

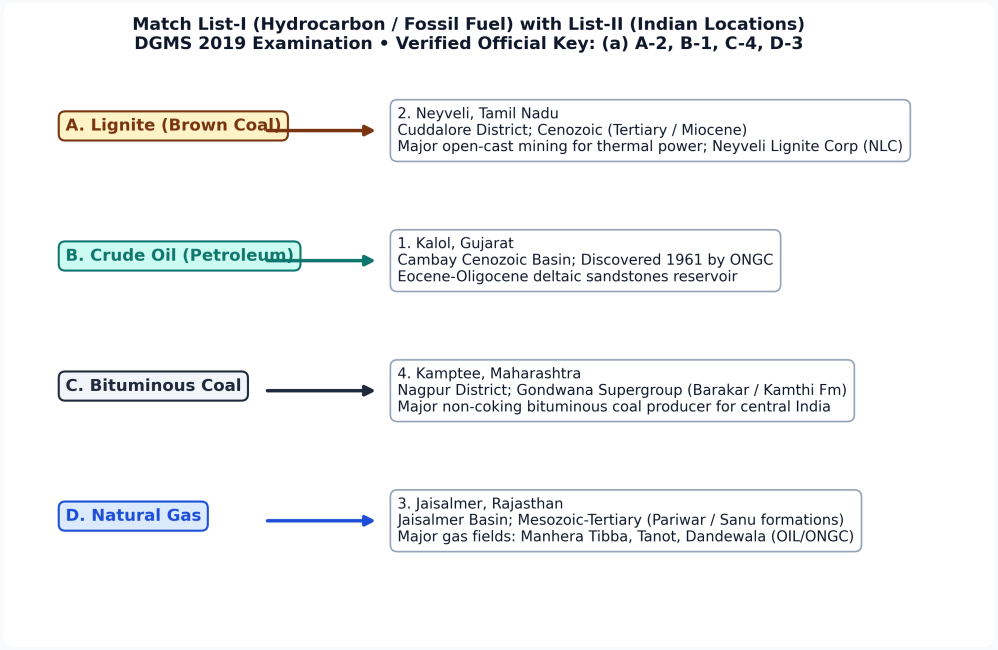


Figure 78.1: Systematic correlation between List-I energy resources and List-II Indian geographical locations with geological and stratigraphic provenance (Verified Official Key: (a) A-2, B-1, C-4, D-3).

2. Detailed Geological & Stratigraphic Provenance of Each Field

Deposit (List-I)	Location (List-II)	Sedimentary Basin	Geological Age & Stratigraphic Unit	Key Geological Characteristics
A. Lignite	**2. Neyveli** (Tamil Nadu)	Cauvery Coastal Basin	Tertiary / Miocene (Cuddalore Sandstone Formation)	Seams up to 20 m thick; high moisture (45–50%), low ash (~6%), calorific value ~2800 kcal/kg; mined by open-cast bucket-wheel excavators.
B. Crude Oil	**1. Kalol** (Gujarat)	Cambay Basin (North Block)	Cenozoic / Middle Eocene (Kalol Formation)	Extensional rift basin; interbedded sandstones, silts, and coal bands; deltaic reservoir sandstones (K-III to K-X pay zones).
C. Bituminous Coal	**4. Kamptee** (Maharashtra)	Pranhita-Godavari / Wardha Valley	Lower Gondwana / Early Permian (Barakar &	Non-coking thermal bituminous coal; interbedded

Deposit (List-I)	Location (List-II)	Sedimentary Basin	Geological Age & Stratigraphic Unit	Key Geological Characteristics
			Kamthi Formations)	with fluvial sandstones and shales; thick coal seams in graben/half-graben fault troughs.
D. Natural Gas	**3. Jaisalmer** (Rajasthan)	Jaisalmer Basin (Shelf Basin)	Mesozoic–Cenozoic (Cretaceous Pariwar & Paleocene-Eocene Sanu Formations)	Sweet gas (predominantly methane with minor nitrogen); reservoirs in shallow marine to fluvial sandstones and shelf carbonates.

3. Evaluation of Match Combinations

- **(a) A-2, B-1, C-4, D-3: CORRECT.** All four geographic, geological, and commodity correlations are verified against official GSI, ONGC, and Coal India records.
- **(b) A-1, B-2, C-3, D-4: Incorrect.** Kalol is an oilfield, not a lignite deposit; Neyveli is a lignite field, not an oilfield.
- **(c) A-2, B-4, C-3, D-1: Incorrect.** Kamptee is a coalfield, not an oilfield; Kalol is an oilfield, not a gas field.
- **(d) A-3, B-1, C-4, D-2: Incorrect.** Jaisalmer is a gas field, not a lignite deposit; Neyveli produces lignite, not natural gas.

> **Key Takeaway:** Neyveli = Lignite; Kalol = Oil; Kamptee = Bituminous Coal; Jaisalmer = Natural Gas. Hence, **(a) A-2, B-1, C-4, D-3** is the exact answer.

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